EXERCISE 6 - STATISTICAL DEPENDENCE AND ITS IMPACT

Consider two discrete random variables

$$\Omega_x = \{x1, x2\}, \Omega_y = \{y1, y2\}.$$

Construct two real-world examples:

1. Example A: X and Y are statistically independent.
2. Example B: X and Y are statistically dependent.
3. For each example:
   - **I.** Clearly explain the real-world meaning of X and Y.
   - **II.** Define a realistic joint pmf p(x, y) and represent it as a 2 × 2 table.
   - **III.** Compute the marginal pmfs p(x) and p(y).
   - **IV.** Compute H(X), H(Y), H(X, Y).

Example A: Statistically Independent Random Variables

1. Real-World Meaning of X and Y

- X **(Commuter Umbrella Status):** Whether a randomly selected commuter in Turin carries an umbrella on an autumn morning.
  - x_1: Carries an umbrella ($P(X = x_1) = 0.4$)
  - x_2: Does not carry an umbrella ($P(X = x_2) = 0.6$)
- Y **(Online Quiz Result):** Whether a student passes a specific online Python programming quiz on the same day.
  - y_1: Passes the quiz ($P(Y = y_1) = 0.7$)
  - y_2: Fails the quiz ($P(Y = y_2) = 0.3$)

Justification: A commuter's choice to carry an umbrella has no physical or causal relationship with a student passing an online coding quiz. Therefore, X and Y are **statistically independent**.

2. Joint PMF $p(x, y)$ as a 2×2 Table

Since X and Y are independent, the joint pmf is given by $p(x, y) = p(x)p(y)$:

X\Y	**y1 (Pass)**	**y2 (Fail)**	**Marginal p(x)**
x_1 **(Umbrella)**	$0.4 \times 0.7 = \mathbf{0.28}$	$0.4 \times 0.3 = \mathbf{0.12}$	0.40
x_2 **(No Umbrella)**	$0.6 \times 0.7 = \mathbf{0.42}$	$0.6 \times 0.3 = \mathbf{0.18}$	0.60
Marginal $p(y)$	0.70	0.30	1.00

3. Marginal PMFs $p(x)$ and $p(y)$

- **Marginal of X:**
  - $p(x_1) = 0.40$
  - $p(x_2) = 0.60$
- **Marginal of Y:**
  - $p(y_1) = 0.70$
  - $p(y_2) = 0.30$

4. Entropies: $H(X)$, $H(Y)$, and $H(X,Y)$

- $H(X)$:

$$H(X) = -\sum_{x} p\,(x) \log_2 p\,(x) = -(0.4 \log_2 0.4 + 0.6 \log_2 0.6) \approx \mathbf{0.9710} \text{ bits}$$

- $H(Y)$:

$$H(Y) = -\sum_{y} p\,(y) \log_2 p\,(y) = -(0.7 \log_2 0.7 + 0.3 \log_2 0.3) \approx \mathbf{0.8813} \text{ bits}$$

- $H(X,Y)$:

$$H(X,Y) = -\sum_{x,y} p\,(x,y) \log_2 p\,(x,y) = H(X) + H(Y) = 0.9710 + 0.8813 = \mathbf{1.8523} \text{ bits}$$

Example B: Statistically Dependent Random Variables

1. Real-World Meaning of X and Y

- X **(Morning Rain Status):** Whether it rains in the morning in the city.
  - x_1: Raining ($P(X = x_1) = 0.30$)
  - x_2: Not raining ($P(X = x_2) = 0.70$)
- Y **(Bakery Hot Soup Sales):** Whether a local bakery sells out of hot soup by noon.
  - y_1: Sells out of soup ($P(Y = y_1) = 0.40$)
  - y_2: Does not sell out of soup ($P(Y = y_2) = 0.60$)

Justification: Rainy mornings strongly increase the customer demand for warm food items like hot soup. Thus, X and Y are **statistically dependent**.

2. Joint PMF $p(x, y)$ as a 2×2 Table

X\Y	y1 (Sells Out)	y2 (Doesn't Sell Out)	Marginal p(x)
x_1 (Rain)	$\mathbf{0.25}$	$\mathbf{0.05}$	0.30
x_2 (No Rain)	$\mathbf{0.15}$	$\mathbf{0.55}$	0.70
Marginal $p(y)$	0.40	0.60	1.00

3. Marginal PMFs $p(x)$ and $p(y)$

1. **Marginal of X (Row sums):**
   - $p(x_1) = 0.25 + 0.05 = 0.30$
   - $p(x_2) = 0.15 + 0.55 = 0.70$
2. **Marginal of Y (Column sums):**
   - $p(y_1) = 0.25 + 0.15 = 0.40$
   - $p(y_2) = 0.05 + 0.55 = 0.60$

4. Entropies: $H(X)$, $H(Y)$, and $H(X, Y)$

- $H(X)$:

$$H(X) = -(0.3 \log_2 0.3 + 0.7 \log_2 0.7) \approx \mathbf{0.8813} \text{ bits}$$

- $H(Y)$:

$$H(Y) = -(0.4 \log_2 0.4 + 0.6 \log_2 0.6) \approx \mathbf{0.9710} \text{ bits}$$

- $H(X, Y)$:

$$H(X, Y) = -(0.25 \log_2 0.25 + 0.05 \log_2 0.05 + 0.15 \log_2 0.15 + 0.55 \log_2 0.55) \approx \mathbf{1.6013} \text{ bits}$$

(Note: $H(X, Y) < H(X) + H(Y)$, illustrating that knowing X reduces uncertainty about Y due to positive mutual information).

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>_ Terminal

› run('Exercise: 6.m')
  ═══ EXAMPLE A: INDEPENDENT ═══
  Marginal p(x): [0.40, 0.60]

  Marginal p(y): [0.70, 0.30]

  H(X) = 0.9710 bits

  H(Y) = 0.8813 bits

  H(X,Y) = 1.8522 bits


  ═══ EXAMPLE B: DEPENDENT ═══
  Marginal p(x): [0.30, 0.70]

  Marginal p(y): [0.40, 0.60]

  H(X) = 0.8813 bits

  H(Y) = 0.9710 bits

  H(X,Y) = 1.6010 bits
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EXERCISE 6 (CONTINUED)

For both real-world examples introduced in the previous slide:

1. Compute the conditional entropies
$$H(X|Y), H(Y|X).$$
2. Verify the chain rules
$$H(X, Y) = H(Y) + H(X|Y) \text{ and } H(X, Y) = H(X) + H(Y|X).$$
3. Compute the mutual information
$$I(X, Y) = H(X) + H(Y) - H(X, Y).$$
4. Compare the two examples and explain the results in terms of the information that observing one variable provides about the other.

In particular, verify that
$$X \text{ and } Y \text{ independent} \Longrightarrow I(X, Y) = 0$$
and compare this result with the statistically dependent case.

Example A: Statistically Independent Random Variables (X and Y)

1. Conditional Entropies $H(X|Y)$ and $H(Y|X)$

For independent random variables, observing one variable provides zero additional information about the other. Therefore, conditional entropy equals the unconditional entropy:

- $H(X|Y) = H(X)$:

$$H(X|Y) = 0.9710 \text{ bits}$$

- $H(Y|X) = H(Y)$:

$$H(Y|X) = 0.8813 \text{ bits}$$

2. Verification of Chain Rules

- **Chain Rule 1 ($H(X,Y) = H(Y) + H(X|Y)$):**

$$H(Y) + H(X|Y) = 0.8813 + 0.9710 = \mathbf{1.8523} \text{ bits} \quad (\text{Matches } H(X,Y))$$

- **Chain Rule 2 ($H(X,Y) = H(X) + H(Y|X)$):**

$$H(X) + H(Y|X) = 0.9710 + 0.8813 = \mathbf{1.8523} \text{ bits} \quad (\text{Matches } H(X,Y))$$

3. Mutual Information $I(X,Y)$

$$I(X,Y) = H(X) + H(Y) - H(X,Y) = 0.9710 + 0.8813 - 1.8523 = \mathbf{0.0000} \text{ bits}$$

Example B: Statistically Dependent Random Variables (X and Y)

1. Conditional Entropies $H(X|Y)$ and $H(Y|X)$

Using the definition $H(X|Y) = H(X,Y) - H(Y)$ and $H(Y|X) = H(X,Y) - H(X)$:

- $H(X|Y)$:

$$H(X|Y) = 1.6013 - 0.9710 = \mathbf{0.6303} \text{ bits}$$

- $H(Y|X)$:

$$H(Y|X) = 1.6013 - 0.8813 = \mathbf{0.7200} \text{ bits}$$

2. Verification of Chain Rules

- **Chain Rule 1 ($H(X,Y) = H(Y) + H(X|Y)$):**

$$H(Y) + H(X|Y) = 0.9710 + 0.6303 = \mathbf{1.6013} \text{ bits} \quad (\text{Matches } H(X,Y))$$

- **Chain Rule 2 ($H(X,Y) = H(X) + H(Y|X)$):**

$$H(X) + H(Y|X) = 0.8813 + 0.7200 = \mathbf{1.6013} \text{ bits} \quad (\text{Matches } H(X,Y))$$

3. Mutual Information $I(X,Y)$

$$I(X,Y) = H(X) + H(Y) - H(X,Y) = 0.8813 + 0.9710 - 1.6013 = \mathbf{0.2510} \text{ bits}$$

Comparison and Interpretation of Results

- **Independent Case (Example A):**
  - $I(X,Y) = 0$, which mathematically verifies that X and Y are completely independent.
  - Observing whether a commuter has an umbrella gives **zero information** about whether a student passes a Python coding quiz ($H(X|Y) = H(X)$).
- **Dependent Case (Example B):**
  - $I(X,Y) = 0.2510$ bits > 0.
  - Observing that it is raining in the morning reduces our uncertainty about the bakery's soup sales. Specifically, our uncertainty drops from $H(Y) = 0.9710$ bits down to $H(Y|X) = 0.7200$ bits, showing that statistical dependence allows one variable to share information about the other.

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Terminal

Type "help" for documentation.
✓ Auto-saved work restored
› run('Exercise: 6 Continued.m')
=== EXAMPLE A: INDEPENDENT ===
H(X|Y) = 0.9710 bits

H(Y|X) = 0.8813 bits

I(X,Y) = 0.0000 bits


=== EXAMPLE B: DEPENDENT ===
H(X|Y) = 0.6301 bits

H(Y|X) = 0.7197 bits

I(X,Y) = 0.2512 bits
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